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Why Won't it Fracture? Investigating the Low Success Rate in the Tigh Gas Formation, North of Oman

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Abstract

The Amin Formation in northern Oman spans two structurally and depositional distinct basins—the Fahud Salt Basin (FSB) and the Ghaba Salt Basin (GSB) with the Triplet Fields forming a notable cluster of fields within the GSB. Despite generally similar reservoir properties and mineralogical makeup, hydraulic fracturing success across the Amin Formation has shown substantial spatial and vertical variability. Stratigraphically, the Amin Formation is subdivided into three principal units arranged from top to bottom as S30, S20, and S10. Particularly, the S30 zone consistently delivers better frac performance than the overlying S20 and S10 intervals. This study investigates the reasons behind such variation by analyzing structural heterogeneity, stress regimes, and sedimentary architecture, with a focus on the geomechanical role of cross-bedding and bedding-parallel anisotropy in enhancing fracture development and propagation.

To understand these variations, an integrated workflow was followed that included gamma-ray profiling of Amin outcrops to delineate stratigraphic subunits, borehole image log interpretation to identify breakout zones and induced fractures, and cross-bedding feature density analysis across wells in both basins. Geomechanical modeling using Kirsch equations was conducted under various stress regimes normal, strike-slip, and reverse faulting to evaluate the impact of stress anisotropy on hoop and radial stress magnitudes. Laboratory and log-derived mechanical properties such as unconfined compressive strength (UCS), cohesion, and friction angle were integrated with structural features to assess failure thresholds. Additionally, radioactive tracer data was cross-referenced with gamma-ray log signatures to validate the relationship between cross-bedding prevalence and effective fracture propagation.

The analysis revealed a clear stratigraphic trend, with cross-bedding density increasing significantly from S10 to S30. These cross-bedded intervals, particularly in S30, showed higher correlation with induced fractures and successful frac stages, as confirmed by radioactive tracer distribution and field image logs. While breakouts were primarily associated with flat, cemented sandstone—indicating compressive stress regimes induced fractures preferentially formed in cross-bedded zones, where mechanical anisotropy likely promotes tensile failure at lower effective stress differentials. Stress modeling demonstrated that under high anisotropy, hoop stress increases while radial stress decreases, setting conditions conducive to fracture initiation, especially in zones with structural heterogeneity. The ability of cross-bedded units to redistribute

stress and facilitate fracture containment was further supported by the observed alignment between image-log fractures and stratigraphic fabric.

This study provides a novel, data-driven interpretation linking sedimentary architecture particularly cross-bedding with hydraulic fracture success in deep, tight formations. By quantifying cross-bedding as a proxy for fracability, this work offers a practical screening tool for future well placement and stimulation design. The findings challenge conventional assumptions that treat homogeneous intervals as mechanically ideal, highlighting instead the beneficial role of heterogeneity in stress perturbation and fracture containment. Operationally, these insights allow for better targeting of fracable intervals, reduced stimulation risk, and improved return on investment in tight gas reservoirs across North Oman

Introduction

In North Oman, Petroleum Development Oman (PDO) is exploring for gas from the Cambro-Ordovician age Haima Supergroup, the Amin reservoir located at depths of over 4,000 m TVD ss. This formation is characterized as a tight sandstone reservoir with porosities ranging from less than 3 to 10% with (ambient) permeabilities ranging from 0.001 to 1 Md (Petroleum Development Oman, 2022).

The Amin Formation, the lowermost unit of the Haima Supergroup, is currently one of the main gas producing stratigraphic intervals in North Oman and is presently responsible for about a good percentage of the total volume produced from the Haima Group (Fig. 1). Since the first discovery of gas and condensate the Amin play has delivered progressively increasing gas and condensate volumes, mainly following significant discoveries in the Ghaba Salt Basin (GSB). During the decade of the 2000s, discoveries in the Fahud Salt Basin (FSB) have further increased the focus on Amin gas in North Oman and encouraged the drilling of steady stream of new exploration and appraisal wells (Petroleum Development Oman, 2022).

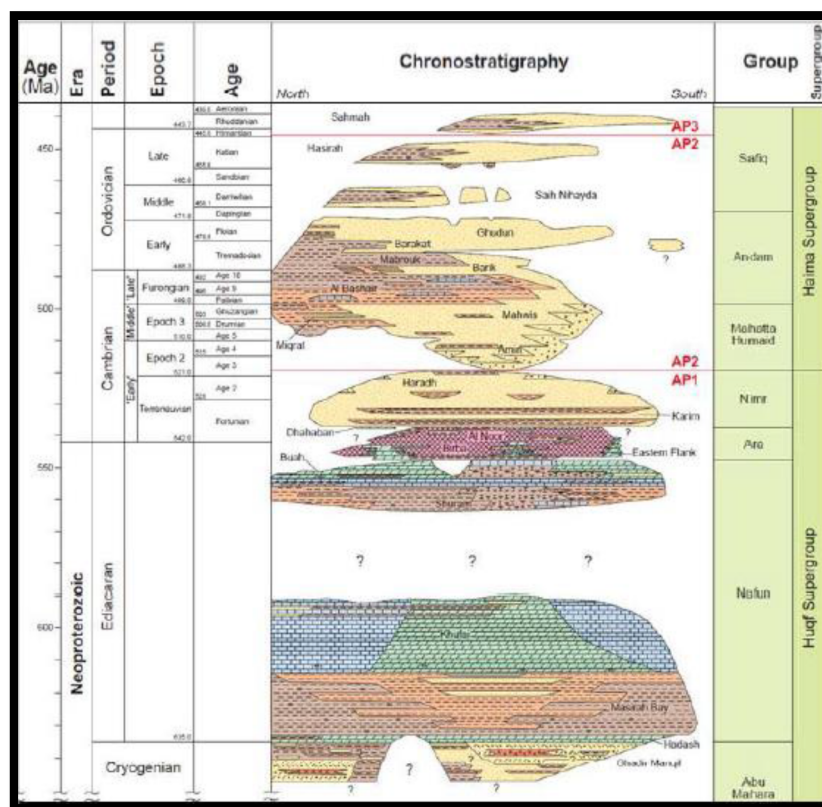


Figure 1—Lithostratigraphy of Oman (Forbas et al. 2010)

The Amin Formation, deposited during the Early Cambrian approximately 520–510 million years ago, accumulated within a semi-arid continental basin variably underlain by salt. Sediment supply was primarily derived from fluvial and alluvial fans draining uplifted basin margins into a broad, low-relief basin center. Over time, significant reworking by aeolian and sabkha processes produced clean, mature sandstones in many intervals. Aeolian dune facies accumulated during drier periods as migrating dunes advanced across the basin floor, while aeolian sand sheets formed from wind-driven ripples in flatter areas. Fluvial channel-fills and alluvial sand sheets record wetter phases with episodic high-energy floods, whereas sabkha and playa deposits reflect lower-energy evaporite-rich conditions and periods of standing water (Petroleum Development Oman, 2022).

Stratigraphically, the Amin has been subdivided into three main intervals S-10, S-20, and S-30 based on characteristic gamma-ray profiles that reflect differences in depositional facies and clay content (Fig. 2). A gamma-ray log profile from an Amin outcrop illustrates this subdivision clearly in the Figure 1, the y-axis shows outcrop height from the Buah at the base up to the Miqrat surface (~130 meters), while the x-axis records total and spectral gamma counts per minute, including total gamma, potassium, uranium, and thorium. The S10 has moderate total gamma and less variability, consistent with clean alluvial sandstones. The S20 shows a serrated gamma pattern with higher peaks, reflecting alternating fluvial and finer grained sabkha and playa deposition. The S30 has consistently lower gamma readings upward, indicating mature, cleaner aeolian sands with reduced clay content, and is locally truncated by the Miqrat Unconformity (Petroleum Development Oman, 2022).

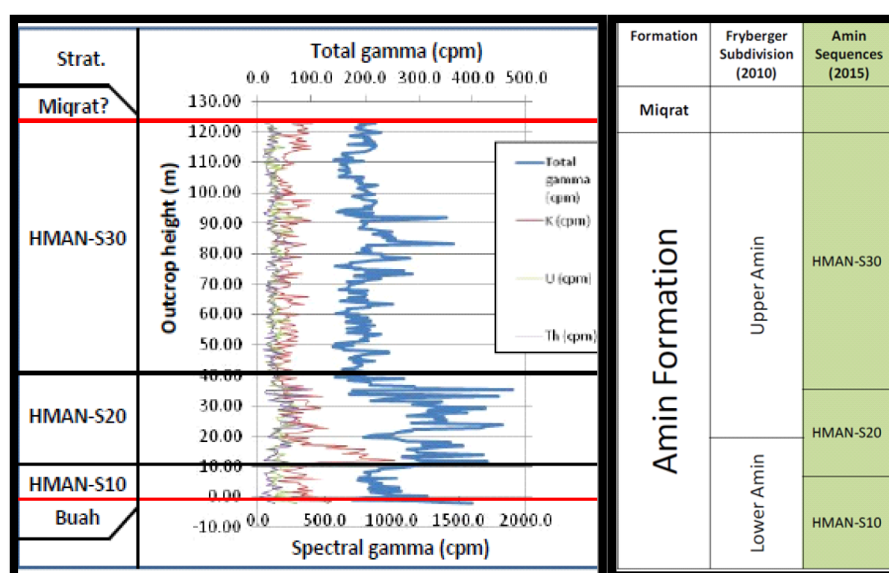


Figure 2—Amin subdivision based on internal understanding (Petroleum Development Oman, 2022)

The integrated regional analysis of the Amin play in the Fahud and Ghaba Salt Basins reveals that three sedimentologically defined stratigraphic units that can be correlated regionally and at field scale (S10, S20 and S30), each with distinct reservoir potential. The uppermost unit (S30) is considered to have the most potential, specifically when dominated by sandstones that have been reworked and/or deposited by aeolian processes (Ghaba Basin), but also where clean fluvial deposits occur (Fahud Basin). Deposition was largely controlled by the overall basin configuration and subsidence regime, with these factors being particularly favorable in the Ghaba Basin allowing thick aeolian influenced deposits with good primary porosities to accumulate, and only minimal post-Amin erosion occurred (Petroleum Development Oman, 2022).

Because the Amin Formation is very tight, hydraulic fracturing is essential for production; however, results have been mixed across the basin. S30 generally shows higher frac success compared to S20 and S10, but even wells with similar porosity can have markedly different breakdown pressures and production rates.

A detailed review of well outcomes shows clear spatial patterns:

- **Top right quadrant of the map:** This area, likely corresponding to the Triplet Fields (Extended to Gaba Salt Basin), shows a dense cluster of wells that were successfully fractured. Here, breakdown pressures were reached consistently, fractures propagated effectively, and production improved, reflecting favorable rock quality and operational experience.
- **Central to lower middle part of the map (GSB):** This region contains many wells that either failed to achieve breakdown pressure or did not deliver any production increase after fracturing, potentially linked to the proximity of the Maradi fault.
- **Left side of the map (FSB):** The results here are mixed, with both successful and failed wells.

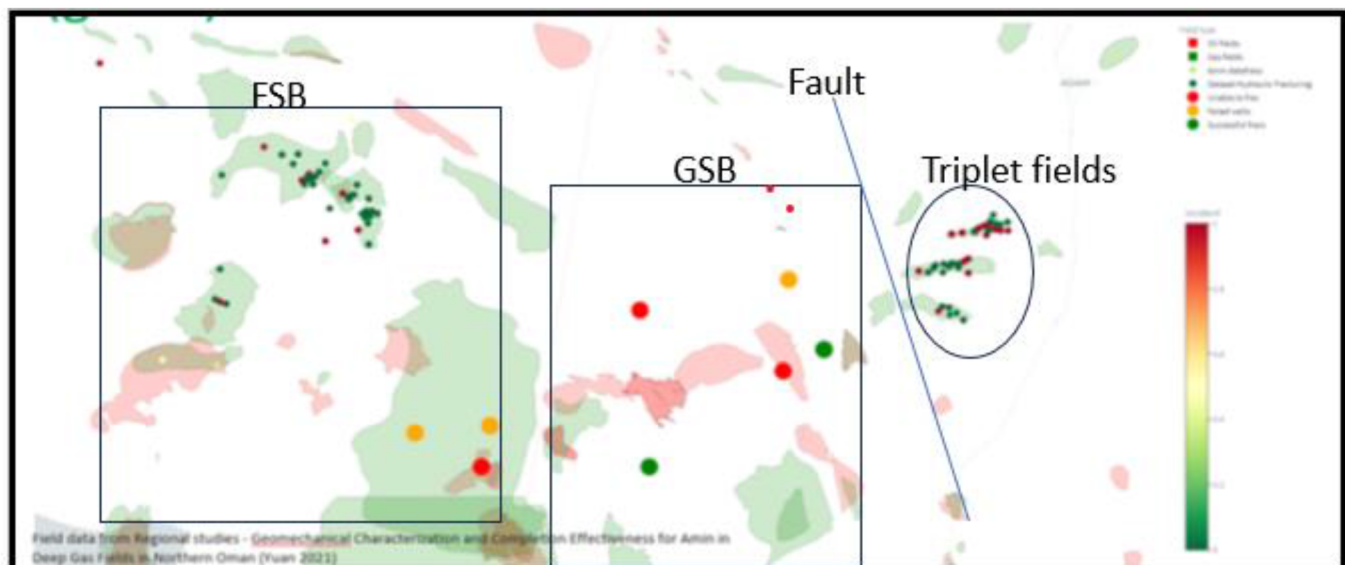


Figure 3—schematic Map showing the spatial distribution of hydraulic fracturing outcomes across the Fahud Salt Basin (FSB), Ghaba Salt Basin (GSB), and Triplet Fields in Northern Oman. Green dots represent successful fracs, while red and orange dots indicate partially fractured or failed treatments. The map also highlights a regional fault separating the GSB from the Triplet Fields. Data adapted from Yuan (2021).

Numerous internal and external studies have examined the Amin Formation from various geological and geomechanics perspectives. However, the specific focus of this study is to investigate and gain a deeper understanding of the markedly low hydraulic fracturing success rates observed in the lower units of the formation—particularly the S-20 and S-10 intervals—compared to the upper S-30 unit. In addition, this work aims to enhance spatial understanding of breakdown pressure variability across the S-30, S-20, and S-10 intervals within the Amin Formation.

Background and Theoretical Framework

Breakouts and induced fractures interpreted from borehole image logs were evaluated to constrain the local in-situ stress regime within the Amin Formation. Breakouts typically form when the compressive hoop stress (σ_θ) at the borehole wall exceeds the compressive strength of the rock, while induced fractures occur when tensile conditions are reached and exceed the rock's tensile strength (Zoback, 2007). For a vertical borehole in a normal or strike-slip stress regime, the hoop stress is expressed by the classical Kirsch solution:

$$\sigma_\theta = (S_{Hmax} + S_{Hmin}) - 2(S_{Hmax} - S_{Hmin})\cos(2\theta) - P_m \quad (\text{Eq 1})$$

where σ_θ is the tangential stress at the borehole wall, SH_{max} and Sh_{min} are the maximum and minimum horizontal stresses, θ is the angle around the borehole, and P_m is the mud pressure (Aadnøy, 1990). Breakouts develop where σ_θ is maximized (i.e., parallel to SH_{max}), while tensile fractures appear where σ_θ is minimized (perpendicular to SH_{max}). The condition for tensile failure is:

$$\sigma_\theta, \min = 3Sh_{min} - SH_{max} - P_m < -T_0 \quad (\text{Eq 2})$$

Where T_0 is the tensile strength of the formation. The plotted graph illustrates the influence of horizontal stress anisotropy on the wellbore stress response. The x-axis represents the anisotropy ratio, defined as the ratio of the maximum to minimum horizontal stresses (SH_{max}/Sh_{min}), while the y-axis shows the normalized hoop (tangential) and radial stresses around the wellbore.

The blue curve corresponds to the normalized hoop stress, which increases progressively with higher anisotropy ratios. This trend indicates that as the contrast between SH_{max} and Sh_{min} increases, the tangential stress concentration around the wellbore intensifies, potentially leading to shear failure (e.g., breakout), depending on the formation's strength.

Conversely, the orange curve represents the normalized radial stress, which decreases with increasing anisotropy. At high anisotropy ratios, the radial stress approaches near-zero values, reflecting a growing susceptibility of the formation to tensile failure.

This stress distribution behavior underscores the critical role of horizontal stress anisotropy in controlling wellbore failure mechanisms.

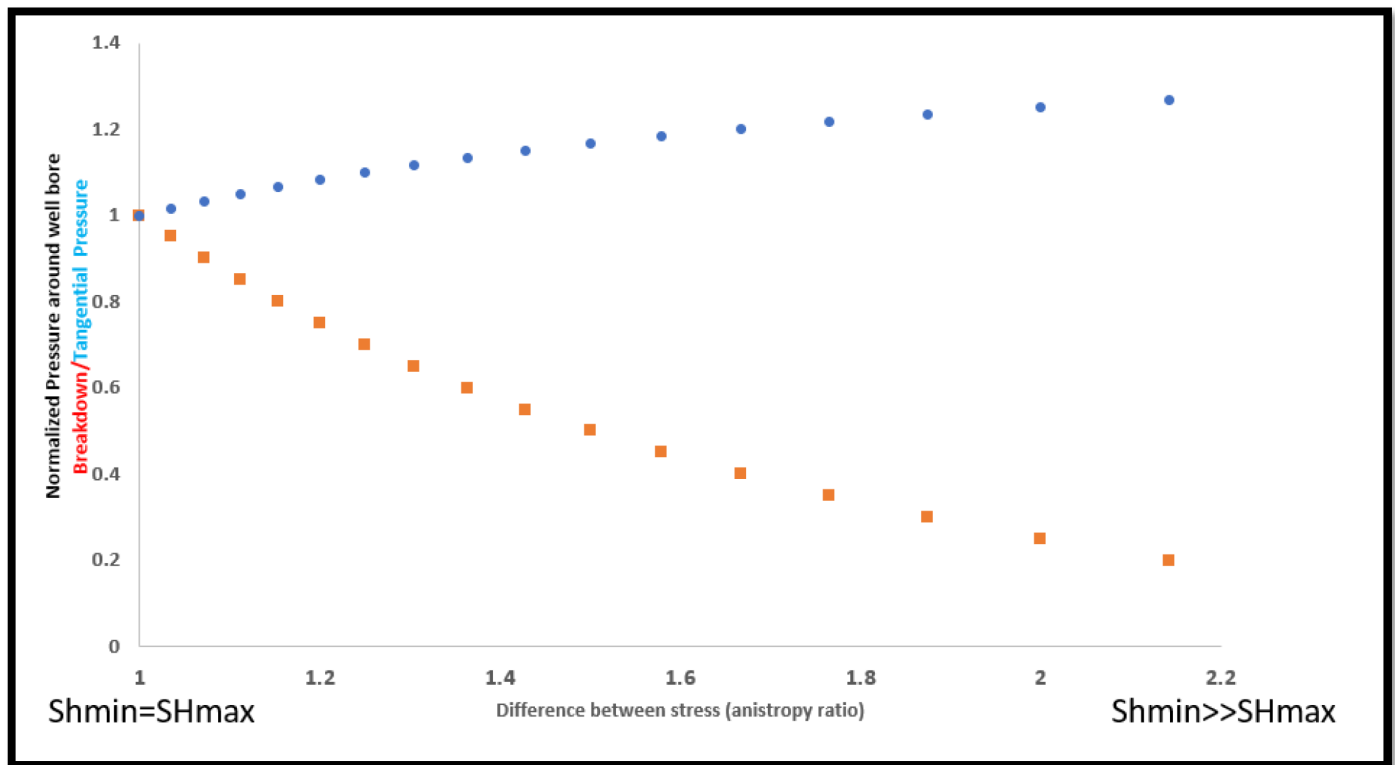


Figure 5—Variation of normalized breakdown pressure (blue) and tangential stress excess (orange) as a function of horizontal stress anisotropy ratio (SH_{max}/Sh_{min}). As the stress difference increases, the tangential stress excess reduces, implying lower tendency for tensile-induced fractures, while breakdown pressure increases, indicating a higher likelihood for shear failure (breakout).

Cross bedding

In geology, cross-bedding also referred to as cross-stratification is a key sedimentary structure defined by inclined laminae that intersect the main horizontal bedding. These sloping layers, called foresets, develop

through the deposition of sediments on the down-current side of bedforms like ripples, dunes, or bars. As these bedforms migrate due to wind or water flow, sediment cascades down the lee slope, producing foreset beds that dip in the direction of the paleocurrent. This structural pattern captures important details about the environment of deposition, including flow direction, energy levels, and transport mechanisms ([Wikipedia, 2024](#)).

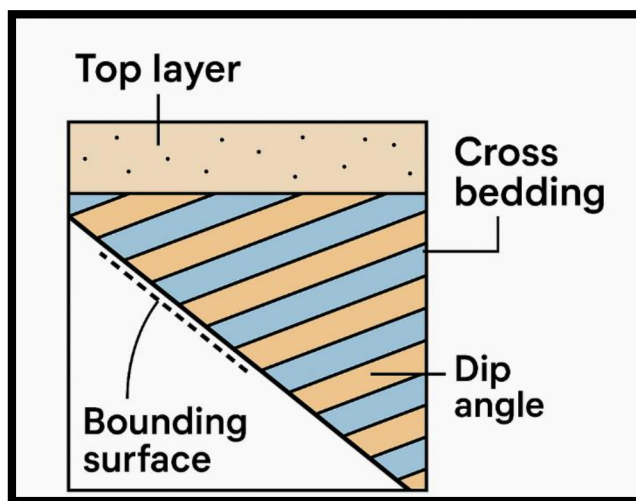


Figure 6—Schematic illustration of cross bedding structure. The diagram highlights key components including the top layer, inclined cross-bedded laminae, dip angle, and bounding surface

The scale and geometry of cross-bedding vary significantly depending on the depositional setting. Aeolian environments typically produce large-scale planar cross-bedding, while fluvial and shallow marine systems often show smaller-scale trough cross-bedding. These foresets are typically separated by bounding surfaces that represent previous crests or erosional truncations from changing flow regimes. Identifying and measuring the dip direction of foresets allows geologists to reconstruct ancient flow directions (paleocurrents), which is critical in regional sedimentological studies. [Figure 7](#) shows the example of large scale of cross bedding in the field and [Figure 8](#) show how such bedding appears in borehole image.



Figure 7—Outcrop view of large-scale cross-bedding in the Amin Formation. A person is included for scale

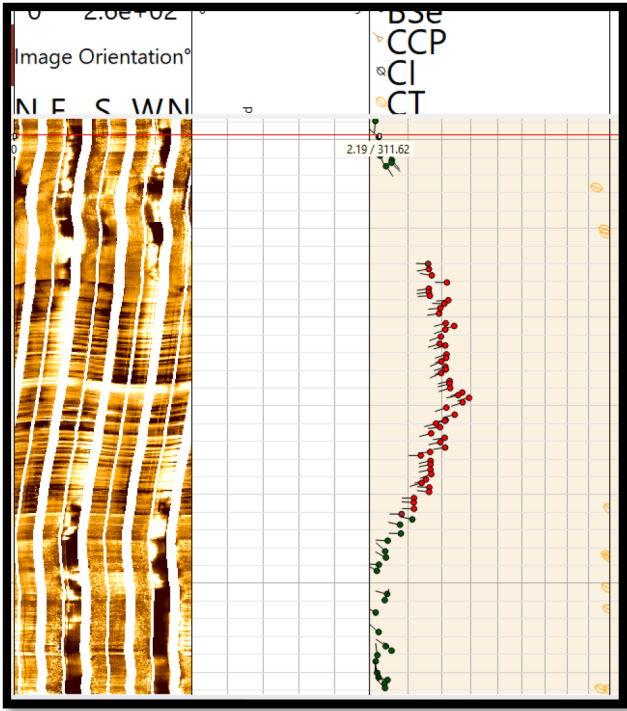


Figure 8—Borehole image log illustrating well-developed cross-bedding in the Amin Formation

The bar chart presents the density of cross-bedded features (features/meter) across three stratigraphic zones within the Amin Formation—S10, S20, and S30 among 12 wells representing both basins. A clear upward trend in cross-bedding density is observed with stratigraphic depth Zone S10 exhibits the lowest cross-bedding density at approximately 4.35 features/m, suggesting relatively less depositional heterogeneity or fewer preserved high-angle bedding structures in this interval. Zone S20 shows a moderate increase, with cross-bedding density around 5.39 features/m, indicating slightly more complex bedding architecture or depositional energy compared to S10. Zone S30 stands out with a significantly higher density of 11.26 features/m, more than double that of S10. This indicates intense cross-bedding development, which may correspond to high-energy depositional environments such as fluvial or aeolian settings.

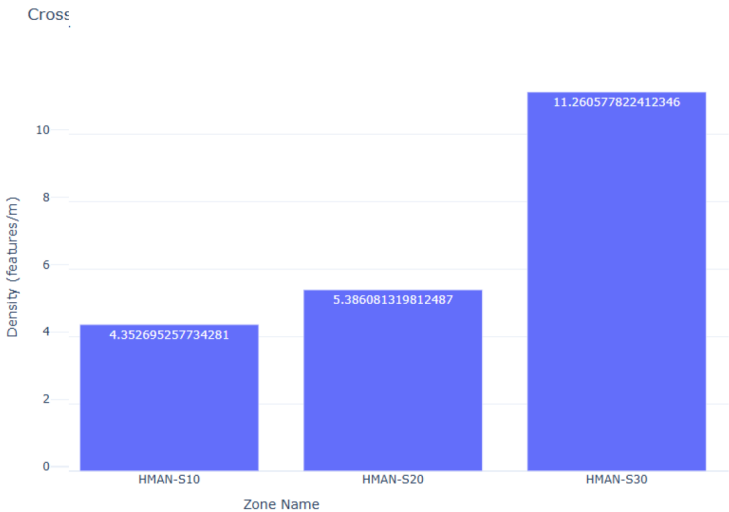


Figure 9—Cross-bedding feature density across stratigraphic zones (S10, S20, S30) through basins

Deep look into Borehole images (BHI)

From multiple wells across the study area spanning two distinct basins. The key features evaluated in this study include cross-bedding, borehole breakouts, and drilling-induced fractures. Figure 11 presents an example from one representative well where all three features are clearly observed.

Statistical analysis across the Amin Formation subunits indicates that these features are present in all intervals, albeit with varying frequency and intensity. Notably, the S30 subunit exhibits the highest density of both induced fractures and breakouts, suggesting a higher degree of stress perturbation or structural complexity. In contrast, the S10 and S20 subunits show a relatively lower occurrence and density of these features, implying a more stable or homogeneous stress environment in those intervals.

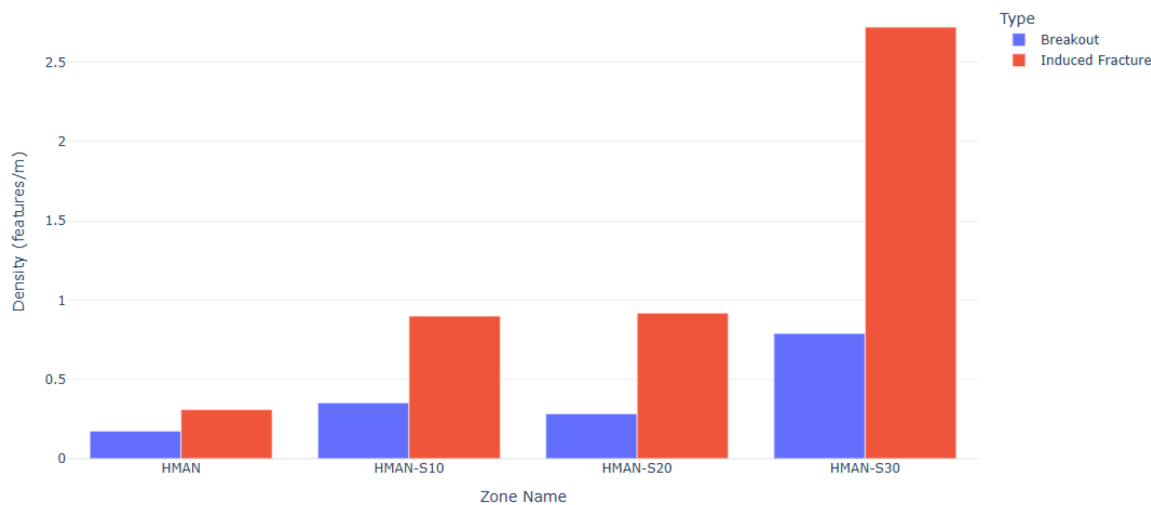


Figure 10—Comparison of borehole breakout and induced fracture densities across different zones of the Amin formation

Figure 11 displays a detailed segment of the borehole image log from Well-XX, highlighting key geological and geomechanical features relevant to this study. The central panel clearly shows well-defined cross-bedding, characterized by inclined sedimentary laminae intersecting the vertical borehole wall at variable angles. These dipping beds reflect changes in depositional energy and paleo-current orientation, commonly associated with high-energy environments such as fluvial channels or tidal sandbars, where inclined foresets are preserved during sedimentation.

Adjacent to the cross-bedded interval—both above and below—are prominent borehole breakout zones, annotated with red circles to the right of the image. These zones are oriented in the NW–SE direction, consistent with the regional orientation of the minimum horizontal stress (Sh_{min}). The presence of these breakouts indicates that localized stress concentrations at the borehole wall exceeded the compressive strength of the formation, resulting in shear failure.

Notably, the cross-bedded interval itself shows a marked reduction or absence of breakouts. This may be attributed to the mechanical heterogeneity introduced by the alternating bed angles, which could enhance stress redistribution and bridging capacity within the rock matrix. Such structural complexity may effectively dampen localized stress concentrations, thereby reducing the likelihood of borehole failure within cross-bedded zones.

In contrast, Figure 12 highlights a cross-bedded interval characterized by steeply dipping laminae; however, unlike the previous example, a drilling-induced tensile fracture is clearly observed within the cross-bedded section itself. This fracture is oriented parallel to the maximum horizontal stress (SH_{max}) direction and is identified on the Formation MicroImager (FMI) log as a planar conductive feature transecting the sedimentary fabric. Its presence within the cross-bedding suggests that, under certain

conditions, the mechanical anisotropy associated with inclined laminae may locally reduce the hoop stress perpendicular to bedding, thereby facilitating tensile failure along SHmax-aligned planes.

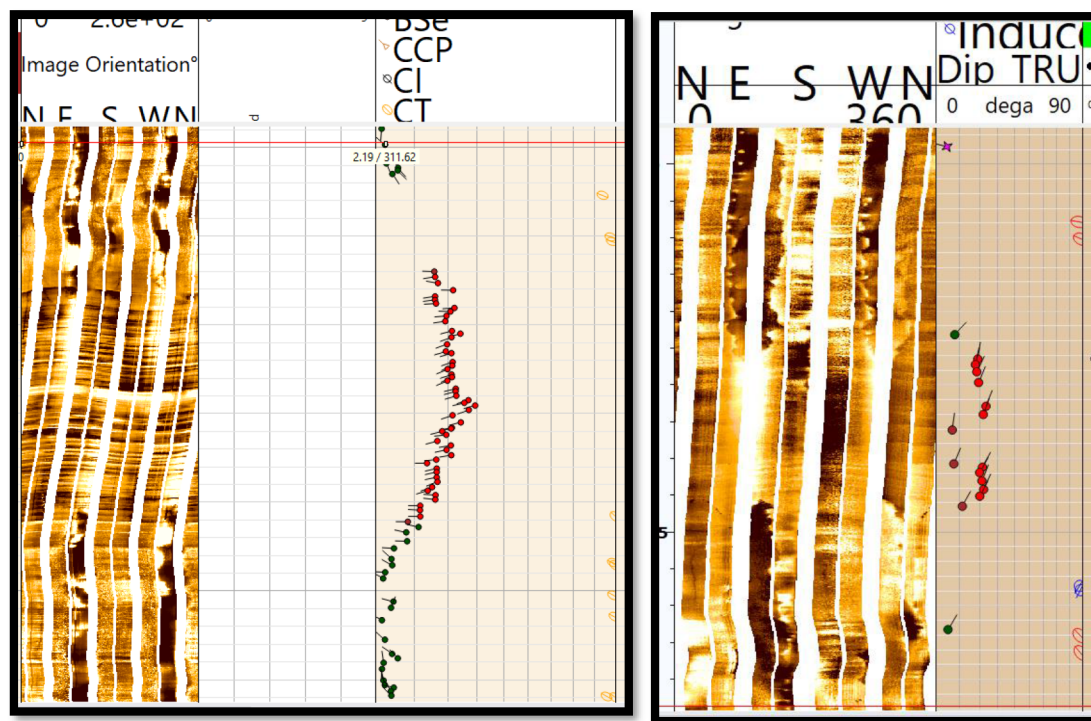


Figure 11—Borehole image and caliper data from Well-XX showing a zone of cross-bedding in the middle interval, characterized by inclined sedimentary structures with varying dip angles. Breakout features are evident above and below the cross-bedded zone

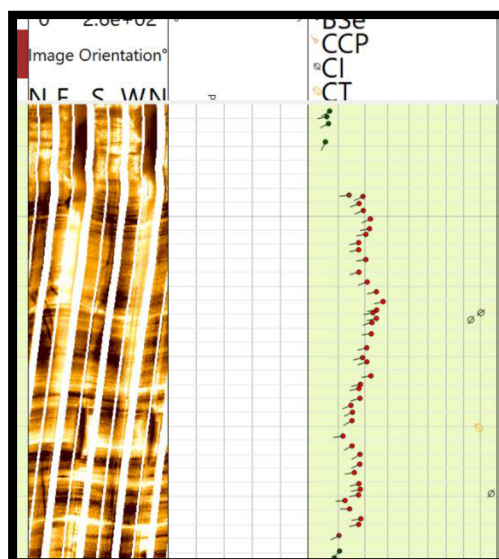


Figure 12—FMI image capturing steeply dipping cross bedding with a clear induced tensile fracture (dark conductive feature) cutting across the bedding in the central zone.

Stress Regime scenarios

In Well-XX, which penetrates the Amin Formation across the S30, S20, and S10 subunits, both borehole breakouts and drilling-induced fractures were observed, despite the use of a consistent mud weight of approximately 14.7 kPa/m. Notably, this mud weight exceeds the estimated pore pressure range of 13.7–

14.0 kPa/m. To investigate this apparent contradiction, three geomechanical stress regime scenarios normal faulting, strike-slip, and reverse faulting and were modeled using Kirsch's equations (Eq. 1 & 2) to evaluate the wellbore stress response. In all scenarios, the vertical stress (S_v) was held constant at 24.5 kPa/m, and minimum horizontal stress (S_{hmin}) was constrained using fracture closure pressure measurements, estimated between 19–20 kPa/m in normal and strike-slip settings.

Case 1: Normal Faulting Stress Regime

Under a normal faulting regime, the stress configuration assumes that S_{Hmax} lies between S_v and S_{hmin} . The modeled stress distribution is shown in Figure 13A. In the high-pressure simulation, the tangential (hoop) stress dominates the stress field, remaining higher than both axial and radial stresses over a pressure range of 15–25 kPa/m. This stress hierarchy typically supports shear-induced breakout development. However, a critical consideration is the high rock strength. Laboratory testing and empirical data suggest an unconfined compressive strength (UCS) of 180–200 MPa for the encountered formations, significantly higher than the modeled tangential stresses making classical shear breakout failure unlikely unless local weaknesses such as bedding anisotropy or pre-existing fractures are present. Interestingly, the model shows that tangential stress transitions into tension (negative values) at pressures above ~24 kPa/m, which is inconsistent with field observations of induced fractures forming as early as 15 kPa/m. This discrepancy may be explained by mechanical layering or natural discontinuities (e.g., cross-bedding) that facilitate tensile failure at lower-than-predicted pressures. Another possibility is the development of en echelon or mixed-mode fractures due to local anisotropy, although this remains speculative.

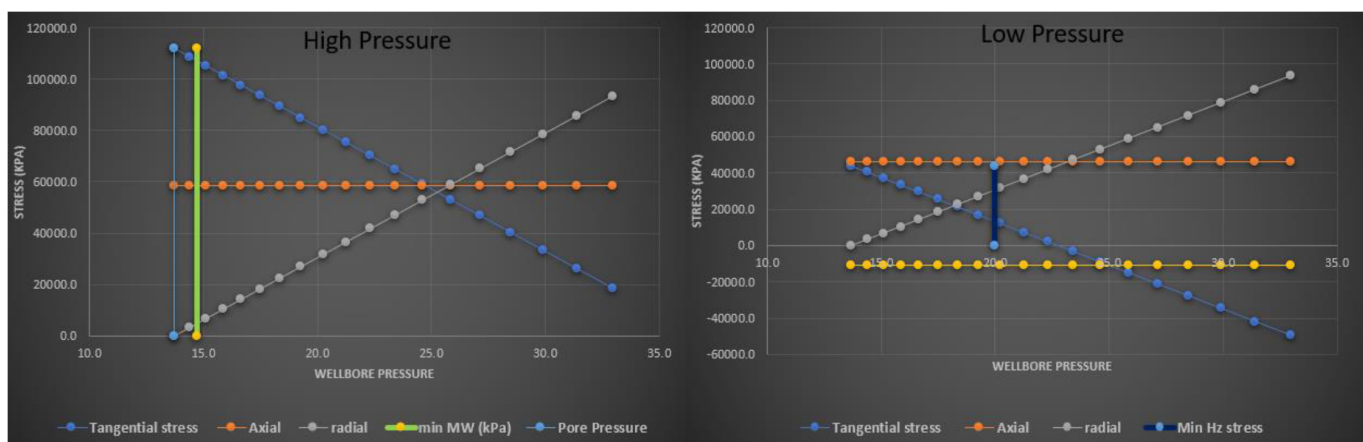


Figure 13—Modeled tangential, axial, and radial stresses around the wellbore under a normal faulting stress regime

Case 2: Strike Slip Stress Regime

In the strike-slip regime, S_{Hmax} is estimated at ~27 kPa/m, while S_{hmin} remains at ~20 kPa/m. This 7 kPa/m differential amplifies the circumferential stress concentration around the wellbore, as modeled by Kirsch's equations (Figure 14). Under high-pressure conditions, tangential stress reaches up to ~160 MPa. Although significant, this remains below the UCS of 180–200 MPa, implying that shear breakout may be limited or delayed due to high rock strength. Conversely, the low-pressure model indicates that tangential stress may enter the tensile regime at mud weights around 20 kPa/m or lower, where it falls below the tensile strength (assumed 8–10 MPa). This aligns with field observations of induced fractures at 14.7 kPa/m, suggesting that tensile failure is plausible even in strong formations, particularly on the low-stress side of the borehole. This scenario underscores the potential for localized tensile fracturing, especially where anisotropic fabric or structural heterogeneities exist. It also highlights the critical role of stress contrast in governing failure mechanisms under strike-slip conditions.

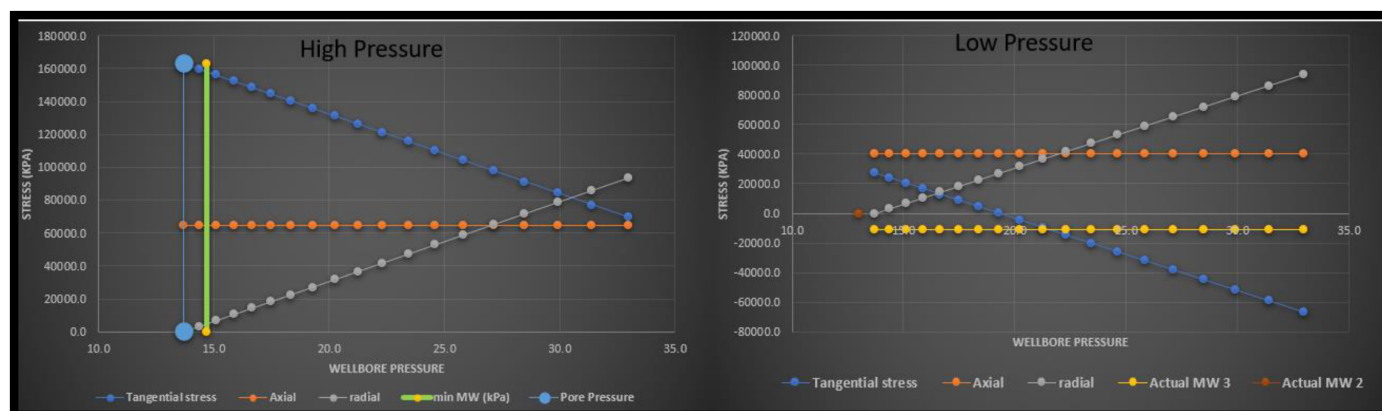


Figure 14—Modeled tangential, axial, and radial stresses around the wellbore under a strike slip faulting stress regime.

Case 2: Reverse Faulting Stress Regime

In a reverse faulting regime, SH_{max} is estimated at ~ 35 kPa/m, and SH_{min} is increased to ~ 27 kPa/m. This substantial stress contrast leads to exceptionally high tangential stresses, reaching over 250 MPa under high-pressure conditions (Figure 15). These values exceed the UCS of the formation, making shear-induced breakout highly probable—even in mechanically competent rocks. However, on the low-pressure side, the analysis shows that mud weights above 30–32 kPa/m would be required to induce tensile failure, well beyond the field-applied mud weight of 14.7 kPa/m. This suggests that induced tensile fractures are unlikely under reverse stress conditions at the applied pressures. In practical terms, this regime promotes shear failure (breakout) while suppressing tensile fracturing, unless extreme overpressure is applied. Any observed fractures under these conditions are more likely due to stress concentration at mechanical discontinuities than classic tensile failure.

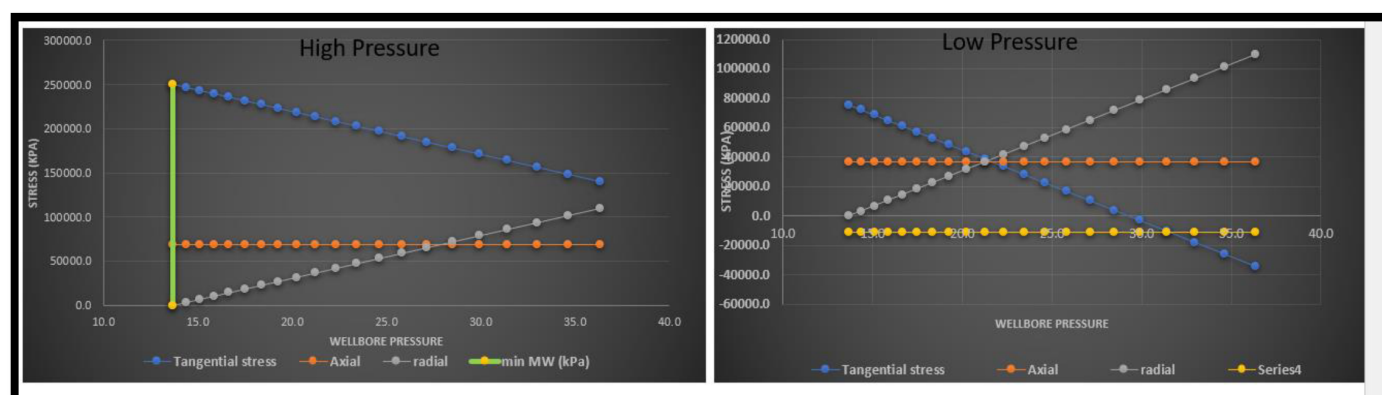


Figure 15—Modeled tangential, axial, and radial stresses around the wellbore under a reverse faulting stress regime.

Discussion

To investigate the relatively low hydraulic fracturing success rate observed in the Amin Formation within the Ghaba Salt Basin (GSB) compared to other basins—particularly the Fahud Salt Basin (FSB) a comparative geomechanical and geological analysis was conducted. The focus was placed on understanding why the lower stratigraphic units of the Amin Formation exhibit reduced stimulation effectiveness. The investigation began with an evaluation of petrophysical and mechanical properties, including rock strength, porosity, mineralogical composition, and reservoir pressure regimes. Additionally, borehole image logs from representative wells in both basins were assessed.

Two dominant sedimentary facies were consistently identified across the Amin Formation in both the FSB and GSB: (i) cemented, horizontally-bedded sandstone, and (ii) cross-bedded sandstone with bedding dips ranging from 20° to 40°. Despite geographic variability, these facies are common throughout the formation. Importantly, borehole image logs revealed the frequent occurrence of borehole breakouts and drilling-induced tensile fractures across wells.

Laboratory-derived geomechanical properties indicated similar rock strength across both basins, with cohesion values ranging between 20–30 MPa and friction angles of 45°–48°. The presence of breakouts in the cemented sandstone units suggests a high differential in horizontal stresses, consistent with either a strike-slip regime—where the maximum horizontal stress (S_{Hmax}) exceeds the vertical stress (SV) or a thrust fault regime where the minimum horizontal stress (S_{Hmin}) approaches or surpasses SV. For context, assuming a vertical stress gradient of ~24.5 kPa/m, measured S_{Hmin} values range between 18–21 kPa/m in strike-slip settings, whereas in a thrust regime, S_{Hmin} values may exceed 24.5 kPa/m.

In contrast, induced tensile fractures, which require a lower horizontal stress differential to initiate, were predominantly observed within cross-bedded sandstone intervals or along bedding planes between cross-bedded layers. This suggests localized stress relaxation in these zones. Importantly, all breakout features were confined to the cemented, horizontally-bedded sandstones and did not propagate extensively, likely due to the inherent strength and cohesion of the rock.

From a fracturing effectiveness perspective, field data revealed that the S30 subunit of the Amin Formation—characterized by abundant cross-bedded sandstone consistently exhibited the highest fracture stimulation success across both basins. This observation is supported by radioactive tracer analysis (e.g. IRF), which indicated enhanced fluid penetration and tracer retention in intervals with pronounced cross-bedding, further corroborated by elevated gamma-ray responses in these zones.

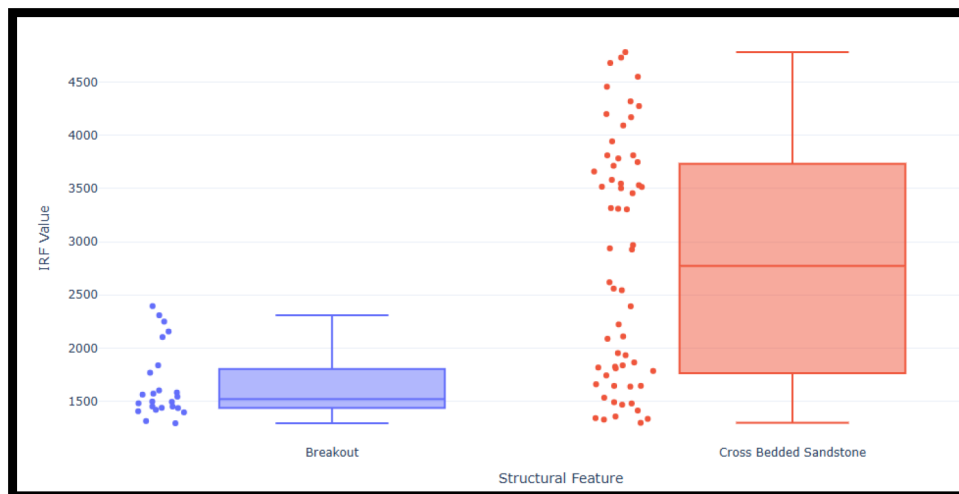


Figure 16—Box plot comparing the type of radioactive tracer called IRF values associated with breakout zones and cross-bedded sandstone intervals. Cross-bedded sandstone shows a significantly broader range and higher median IRF values compared to breakout features

The observed relationship between cross-bedding and fracture propagation may be attributed to stress heterogeneity induced by sedimentary anisotropy. Bedding-parallel weakness planes likely reduce hoop stress perpendicular to the bedding orientation, facilitating tensile failure and enhancing fracture initiation in otherwise tight, high-strength formations. From a reservoir geomechanics perspective, such bedding-induced stress perturbations can be critical in promoting fracture complexity and connectivity.

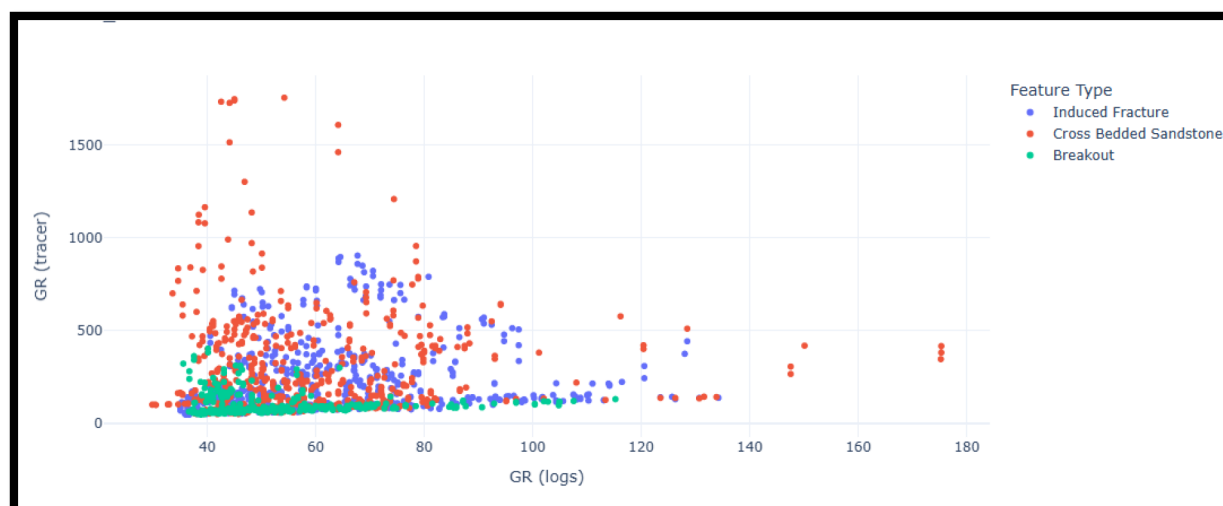


Figure 17—Cross plot of gamma ray (GR) values from logs versus feature counts for different structural features. Induced fractures (red) and cross-bedded sandstone (blue) are more frequently associated with moderate GR values (40-80 API), while breakouts (green) tend to occur at lower GR and feature density ranges

Conclusion and Recommendation

This study highlighted the critical influence of sedimentary structures particularly cross-bedding on fracture initiation and propagation within the tight sandstone reservoir of the Amin Formation. Despite uniformity in lab-measured mechanical strength across wells in both basins, distinct differences in frac outcomes were observed. Breakouts were typically associated with flat, cemented sandstones under high-stress regimes, suggesting significant stress anisotropy, potentially including reverse faulting conditions where the minimum horizontal stress exceeds the overburden. In contrast, induced fractures were consistently found in cross-bedded intervals, where structural heterogeneity likely promotes stress redistribution and facilitates tensile failure at lower-than-expected pressures. The S30 zone, which exhibits the highest density of cross-bedding, was most fracable and showed consistent tracer uptake, reinforcing the role of depositional architecture in governing stimulation response. This integration of geomechanical modeling, borehole image interpretation, and tracer validation provides a comprehensive explanation for spatial and stratigraphic variability in frac performance across the Amin.

Building on these insights, a new predictive approach is recommended for optimizing hydraulic stimulation in tight gas reservoirs of North Oman. Cross-bedding density can be used as a proxy indicator for mechanical anisotropy and fracture-favorability, especially in the S30 interval. Future stimulation strategies should prioritize targeting cross-bedded zones, informed by FMI logs and GR-tracer overlays, to maximize fracture connectivity and improve well productivity. Additionally, fracture modeling should account for stress rotation effects and bedding-related heterogeneity when estimating breakdown pressures or selecting perforation intervals. Incorporating these findings into field development plans will reduce the risk of unsuccessful fracs, enhance reservoir drainage, and improve return on investment, particularly in complex settings near structural features like the Maradi Fault.

In conclusion, this study highlights the importance of sedimentary fabric specifically cross-bedding in modulating in-situ stress conditions and influencing stimulation outcomes. The orientation, scale, and mechanical properties of cross-bedding should therefore be integral considerations in fracture design and well placement strategies in stratified reservoirs. Understanding these anisotropic effects can improve stimulation efficiency and hydrocarbon recovery in geomechanically complex formations such as the Amin.

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